

Cherry Mittal

312-972-5734 | cherrymittal998@gmail.com | [linkedin/cherry-mittal](https://www.linkedin.com/in/cherry-mittal) | [github/cherry-mittal](https://github.com/cherry-mittal) | cherry-mittal.github.io | Chicago, IL

Professional Summary

AI/ML Engineer with 6+ years in software engineering, the last 4 building and shipping production LLM/ML systems. Architects RAG pipelines, agentic LangGraph workflows, evaluation frameworks, and self-hosted LLM serving (vLLM) under real latency, cost, and data-residency constraints. Healthcare/PHI background brings a strong instinct for guardrails and compliance-aware system design.

Technical Skills

Languages: Python, SQL, PostgreSQL, Apex, Javascript, HTML, CSS

Generative AI & LLMs: GPT-4, Llama-3, BERT, PaliGemma, RAG, Prompt Engineering, Structured Outputs, Claude API, AI Agents, Agentic Workflows, Context Engineering

Agentic & Orchestration: LangGraph, LangChain, Human-in-the-Loop Workflows, Tool Calling

Retrieval & Vector Search: RAG, Semantic Search, Embeddings, BGE-M3, Retrieval Optimization, Re-ranking

Model Training: PyTorch, Keras, TensorFlow, Hugging Face, Transformers, LoRA/QLoRA/PEFT, ONNX, Supervised Fine-Tuning

LLM Serving & Inference: vLLM, GPU Inference, PagedAttention, Inference Optimization, Latency/Throughput & Cost Optimization

Evaluation & Guardrails: Golden Datasets Evaluation, Recall@k, Ragas, Prompt Injection Defense, PHI Redaction, Safety Guardrails, Hallucination Detection, A/B Testing

Tools & Technologies: AWS (Sagemaker, Bedrock), Azure, Salesforce Cloud, FastAPI, REST/GraphQL APIs, Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Vector Databases (Pinecone, pgvector, ChromaDB), Claude Code, Dynatrace

ML: Scikit-learn, NLP, Classification, Regression, Feature Engineering, Model Evaluation, Deep Learning, CNNs, RNNs, LSTMs

Experience

AI/ML Engineer (Contract)

August 2025 – Present

Billion Peoples Technologies

Remote

- **Architected & deployed to production an LLM-powered Text-to-SQL engine** using Python, BGE-M3, and PostgreSQL, automating complex data retrieval for 50+ users and **slashing manual SQL reporting overhead by ~70%**.
- **Designed a schema-aware RAG pipeline** that embedded database metadata and retrieved only the top-K relevant tables per query, improving SQL generation accuracy by 30% while reducing hallucinated joins.
- **Built an agentic self-correction loop with LangGraph:** a stateful retrieval, generation, validation, execution graph with bounded retries that recover failed queries.
- **Built a 500-pair golden-dataset eval**, execution-match scoring and Recall@k to regression-test every pipeline change.
- **Engineered multi-layer guardrails on a FastAPI layer:** Pydantic input validation, SQL parsing, SELECT-only allowlist, identifier validation against retrieved schema, and a read-only PostgreSQL role with statement timeouts & row limits as the hard safety boundary.

Senior Software Engineer

Feb. 2022 – Apr. 2025

Optum (United Health Group)

Bangalore, India

- **Architected CAREVOICE**, a production AI call-intelligence platform processing 10K+ customer calls/month, leveraging AWS event-driven services (S3, SNS, SQS) and self-hosted Llama-3 inference via vLLM, reducing post-call documentation effort by ~40%.
- **Fine-tuned a BERT sentiment-analysis pipeline** on 5K labeled customer utterances with class weighting for the minority escalating class and ONNX export for inference, delivering sub-second agent insights and improving CSAT by ~20%.
- **Developed a supervised ML classifier** (Python/Scikit-learn) to identify high-risk insurance members, enabling targeted outreach for 10K+ users and **reducing case resolution time by ~55%**.
- **Directed a 6-engineer team** to launch a VIP insurance platform, achieving a ~40% reduction in system latency.
- **Mentored junior engineers and drove Agile sprint** execution across ML and AI initiatives, guiding key architecture decisions.

Software Engineer

Jan 2019 – Feb 2022

TEKSystems Global Services

Hyderabad, India

- **Architected Salesforce LWC applications** integrated with 30+ enterprise systems for real-time bi-directional healthcare data flow.
- **Re-engineered backend API architecture from REST to GraphQL**, eliminating data over-fetching and **slashing processing latency by ~50%** for 1,000+ concurrent users.
- **Owned production deployment**, managing CI/CD release cycles and post-release stability for a high-traffic platform.
- Implemented OIDC-based SSO, enabling secure access for 1,000+ users and improving login success rates by ~20%.

Selected Projects

Mini-GPT - Transformer Language Model from Scratch | Python, PyTorch, CUDA

Dec 2025 - Feb 2026

Engineered a custom GPT-style autoregressive model in PyTorch from first principles with multi-head self-attention, pre-norm layer normalization, and masked attention, reaching 1.47 cross-entropy loss on TinyShakespeare. Built the full training pipeline (tokenization, automated loss tracking, text generation) and cut GPU memory utilization by ~25% via vectorized operations.

Multimodal Image-Text Classifier - PaliGemma Fine-Tuning | PyTorch, Hugging Face, LoRA

May 2025 - June 2025

Fine-tuned PaliGemma VLM with LoRA, freezing original weights and training low-rank adapter matrices for domain-specific image-text classification, raising F1 by ~12% while training under 1% of total parameters. Built an end-to-end multimodal pipeline with Hugging Face Accelerate, optimizing data loading and prompt formatting to reduce training wall-clock time by ~35%.

Education & Certifications

B.E., Computer Science — Acropolis Institute of Technology

July 2015 - June 2019

Certifications: Generative AI Nanodegree (Udacity); Salesforce Certified Administrator, Platform Developer I, App Builder

Recognition: Spot Award (Optum, 2022); Diamond Award (Optum, 2023)